

The Anatomy of Σ_t and Σ_t^{-1}

How Diffusion Noise Reshapes the Score of an AR(1) Sequence

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Abstract

We study the matrices Σ_t and Σ_t^{-1} that control the joint score $S(\mathbf{x}, t) = -\Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t)$ of a diffusion model applied to an AR(1) Gaussian sequence. At diffusion time $t = 0$ the precision matrix Σ_0^{-1} is tridiagonal—each frame couples only to its nearest neighbours, the algebraic signature of Markovianity. We derive four closed-form representations of Σ_t^{-1} and show that diffusion noise fills in the tridiagonal zeros through a mechanism we make precise: the inverse of a tridiagonal matrix is a full matrix. We prove that Σ_t and Σ_t^{-1} share the eigenvectors of Σ_0 for all t , and that in this eigenbasis the score decouples into independent scalar denoisers. A 60×60 numerical study confirms all predictions. The analysis cleanly separates two roles: *Gaussianity* enables the decoupling; *Markovianity* determines the specific structure of the eigenvectors and eigenvalues.

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1 The Problem

1.1 Context

A score-based diffusion model generates samples by learning the *score function* $\nabla_{\mathbf{x}} \log P_t(\mathbf{x})$ of progressively noisier versions of a target distribution. When the target is a *sequence* of $K + 1$ correlated frames—such as a video or a time series—the score carries information about how frames influence each other. Our previous note derived the exact joint score for a Gaussian AR(1) sequence:

$$S(\mathbf{x}, t) = -\Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t), \quad (1)$$

where $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I_{K+1}$ is the covariance of the noisy observations and $\Delta_t = 1 - e^{-2t}$.

Equation (1) says something powerful: *the entire structure of the score lives inside the precision matrix* Σ_t^{-1} . The matrix Σ_t^{-1} is the object that tells us which frame influences which, and by how much, when computing the gradient of the log-density. This document is a thorough study of what Σ_t^{-1} looks like, what it means, and how it evolves with diffusion time t .

1.2 The central question

During the meeting of March 18, 2026, Mézard posed the driving question:

“What is the impact of the Markovian process underlying the sequence of frames on the structure of the score?”

At $t = 0$ (no noise), the precision matrix Σ_0^{-1} is **tridiagonal**: each frame couples only to its two immediate neighbours. This is the algebraic encoding of the Markov property. As the diffusion noise t increases, what happens? Does the Markov “fingerprint” persist, degrade gradually, or vanish? In a suitable basis, can we make the problem completely independent?

1.3 The AR(1) model

Definition 1.1 (Stationary AR(1) sequence). The clean frames $\mathbf{a} = (a_0, a_1, \dots, a_K)^\top \in \mathbb{R}^{K+1}$ satisfy

$$a_0 \sim \mathcal{N}(0, \sigma_\infty^2), \quad a_{k+1} = \alpha a_k + \eta_k, \quad \eta_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad (2)$$

with $|\alpha| < 1$ (ensuring stability: the variance does not explode) and $\sigma_\infty^2 = \sigma_\eta^2 / (1 - \alpha^2)$.

The contraction $|\alpha| < 1$ ensures that the chain “forgets” its past exponentially: the correlation between frames j and k is $\text{Cov}(a_j, a_k) = \sigma_\infty^2 \alpha^{|j-k|}$, which decays geometrically with the distance $|j - k|$.

We work in the **stationary regime** ($\sigma_0^2 = \sigma_\infty^2$), so the variance is the same at every frame. This makes the covariance matrix Σ_0 have the same value along each diagonal (a *Toeplitz* structure), which gives clean spectral formulas. All structural results hold outside stationarity; the simplification is for clarity.

The **OU forward process** independently corrupts each frame:

$$x_k = e^{-t} a_k + \sqrt{\Delta_t} z_k, \quad z_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad k = 0, \dots, K. \quad (3)$$

The identity $e^{-2t} + \Delta_t = 1$ ensures energy is conserved: what the signal loses to damping, the noise gains.

2 Σ_0 and Σ_0^{-1} : Dense vs. Tridiagonal

2.1 The covariance matrix Σ_0

Proposition 2.1 (Covariance). *The entries of $\Sigma_0 \in \mathbb{R}^{(K+1) \times (K+1)}$ are*

$$(\Sigma_0)_{jk} = \sigma_\infty^2 \alpha^{|j-k|}. \quad (4)$$

Every entry is nonzero (for $\alpha \neq 0$): *the covariance matrix is dense*. This reflects the fact that every frame is marginally correlated with every other frame. Frame 0 and frame 50 have correlation $\alpha^{50} \approx 0.005$ (for $\alpha = 0.9$)—small, but not zero.

2.2 The precision matrix Σ_0^{-1}

Proposition 2.2 (Tridiagonal precision). *In the stationary case, Σ_0^{-1} is symmetric and tridiagonal:*

$$\Sigma_0^{-1} = \frac{1}{\sigma_\eta^2} \begin{pmatrix} 1 + \alpha^2 & -\alpha & & & \\ -\alpha & 1 + \alpha^2 & \ddots & & \\ & \ddots & \ddots & \ddots & -\alpha \\ & & & -\alpha & 1 \end{pmatrix}. \quad (5)$$

The only nonzero entries are on the main diagonal and the first super/sub-diagonals.

Proof. The joint density factorises by the Markov property:

$$\log P_0(\mathbf{a}) = -\frac{a_0^2}{2\sigma_\infty^2} - \sum_{k=0}^{K-1} \frac{(a_{k+1} - \alpha a_k)^2}{2\sigma_\eta^2} + \text{const.} \quad (6)$$

Each term $(a_{k+1} - \alpha a_k)^2 = a_{k+1}^2 - 2\alpha a_{k+1}a_k + \alpha^2 a_k^2$ contributes to at most three matrix entries: the diagonal at k and $k+1$, and the off-diagonal at $(k, k+1)$. No term ever contains a product $a_j a_k$ with $|j - k| \geq 2$. This is a direct algebraic consequence of the Markov property: the log-density involves only nearest-neighbour interactions.

The last diagonal entry is $1/\sigma_\eta^2$ (not $(1 + \alpha^2)/\sigma_\eta^2$) because frame K appears only in the transition *into* it, never as the source of a transition out. $\square$

What Σ_0^{-1} tells us about each frame

For a Gaussian vector, the precision matrix encodes *conditional* relationships. Entry $(\Sigma_0^{-1})_{jk} = 0$ means: knowing a_j provides no additional information about a_k once you already know all other frames. The tridiagonal structure says: **a_k is conditionally independent of all non-neighbours given its two neighbours a_{k-1} and a_{k+1}** . This is the Markov property written in matrix language.

Concretely, the best prediction of a_k given everything else is:

$$\mathbb{E}[a_k \mid \mathbf{a}_{\setminus k}] = \frac{\alpha}{1 + \alpha^2} (a_{k-1} + a_{k+1}). \quad (7)$$

A simple average of the two neighbours, weighted by $\alpha/(1 + \alpha^2)$. For $\alpha = 0.9$: weight ≈ 0.497 . No other frame contributes.

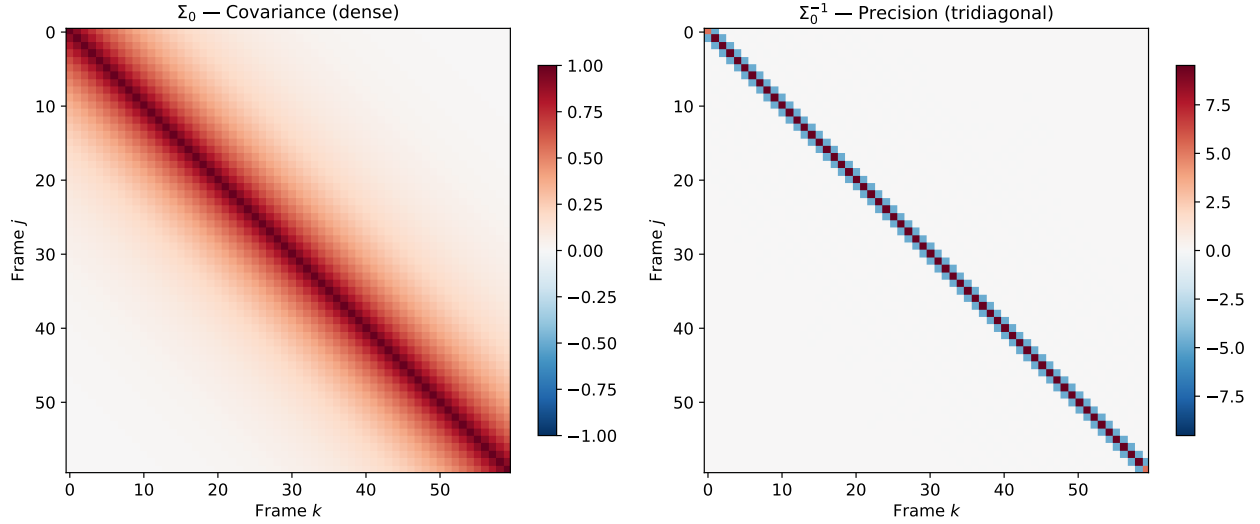


Figure 1: Σ_0 (left) vs. Σ_0^{-1} (right) for 60 frames with $\alpha = 0.9$. The covariance encodes total (marginal) correlations and is dense. The precision encodes direct (conditional) couplings and is tridiagonal: the Markov property in matrix form.

3 Σ_t^{-1} : Four Representations

We now invert $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$ in four different ways, each revealing a different aspect of the answer.

Notation. Define the *signal-to-noise ratio* $\gamma_t = e^{-2t}/\Delta_t$, which goes from ∞ (at $t = 0$, pure signal) through 1 (at $t = \frac{1}{2} \ln 2 \approx 0.347$, equal parts) to 0 (at $t \rightarrow \infty$, pure noise).

The four formulas

$$\Sigma_t^{-1} = \frac{1}{\Delta_t} (I + \gamma_t \Sigma_0)^{-1} \quad (\text{F1: Factored})$$

$$= \Sigma_0^{-1} (e^{-2t} I + \Delta_t \Sigma_0^{-1})^{-1} \quad (\text{F2: Mézard})$$

$$= \frac{1}{\Delta_t} I - \frac{e^{-2t}}{\Delta_t^2} \Sigma_0 (\Sigma_0^{-1} + \gamma_t I)^{-1} \quad (\text{F3: Woodbury})$$

$$= U \operatorname{diag} \left(\frac{1}{e^{-2t} d_m + \Delta_t} \right)_{m=0}^K U^\top \quad (\text{F4: Spectral})$$

where $\Sigma_0 = U \operatorname{diag}(d_0, \dots, d_K) U^\top$ is the eigendecomposition of Σ_0 .

Let us derive and interpret each one.

3.1 F1: Factored form — what controls the overall scale

Proof. Factor Δ_t from Σ_t : $\Sigma_t = \Delta_t (I + (e^{-2t}/\Delta_t) \Sigma_0) = \Delta_t (I + \gamma_t \Sigma_0)$. Invert: $\Sigma_t^{-1} = (1/\Delta_t) (I + \gamma_t \Sigma_0)^{-1}$. $\square$

What F1 tells us

The prefactor $1/\Delta_t$ sets the overall scale: as $t \rightarrow \infty$, $\Delta_t \rightarrow 1$ and the scale is $O(1)$. The “shape” of Σ_t^{-1} is controlled by $(I + \gamma_t \Sigma_0)^{-1}$.

When $\gamma_t \gg 1$ (small t , strong signal): the $\gamma_t \Sigma_0$ term dominates, so $(I + \gamma_t \Sigma_0)^{-1} \approx (\gamma_t \Sigma_0)^{-1} = \Sigma_0^{-1}/\gamma_t$, and $\Sigma_t^{-1} \approx \Sigma_0^{-1}/e^{-2t}$ —tridiagonal.

When $\gamma_t \ll 1$ (large t , strong noise): the I term dominates, so $(I + \gamma_t \Sigma_0)^{-1} \approx I$, and $\Sigma_t^{-1} \approx I/\Delta_t$ —diagonal, fully decoupled.

The parameter γ_t is the “dial” that interpolates between Markov structure and isotropy.

3.2 F2: Mézard’s form — why the zeros fill in

Proof. Left-multiply $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$ by Σ_0^{-1} : $\Sigma_0^{-1}\Sigma_t = e^{-2t}I + \Delta_t \Sigma_0^{-1}$. Therefore $\Sigma_t^{-1} = (\Sigma_0^{-1}\Sigma_t)^{-1}\Sigma_0^{-1} = (e^{-2t}I + \Delta_t \Sigma_0^{-1})^{-1}\Sigma_0^{-1}$. $\square$

What F2 tells us

This formula writes Σ_t^{-1} as a product of two matrices: Σ_0^{-1} (tridiagonal) and $(e^{-2t}I + \Delta_t \Sigma_0^{-1})^{-1}$. The matrix $B = e^{-2t}I + \Delta_t \Sigma_0^{-1}$ is tridiagonal (a diagonal shift of a tridiagonal matrix is still tridiagonal). But—and this is the crux—the **inverse of a tridiagonal matrix is generically full**.

Why? Each entry of B^{-1} involves cofactors of all rows and columns of B (via the adjugate formula $(B^{-1})_{jk} = (-1)^{j+k} M_{kj} / \det B$). No cancellation makes these entries vanish. Alternatively: the eigendecomposition $B^{-1} = \sum_m (1/\beta_m) \mathbf{u}_m \mathbf{u}_m^\top$ sums over all eigenvectors, and since the eigenvectors $\mathbf{u}_m(k)$ are nonzero at every position k , every entry $(B^{-1})_{jk}$ receives contributions from every mode.

The product of a tridiagonal matrix (Σ_0^{-1}) with a full matrix (B^{-1}) is full. This is the mechanism by which diffusion noise fills in the Markov zeros.

3.3 F3: Woodbury form — the noise floor plus a signal correction

The Woodbury identity $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$ with $A = \Delta_t I$, $C = e^{-2t}\Sigma_0$, $U = V = I$ gives F3.

What F3 tells us

Σ_t^{-1} splits into two pieces: a “noise floor” I/Δ_t (diagonal, no coupling) minus a “signal correction” that carries all the inter-frame structure. At large t , the correction vanishes and the noise floor is all that remains.

Moreover, if $\gamma_t < 1/\|\Sigma_0\|$ (which holds at large enough t), we can expand $(I + \gamma_t \Sigma_0)^{-1}$ as a Neumann series:

$$\Sigma_t^{-1} = \frac{1}{\Delta_t} \sum_{n=0}^{\infty} (-\gamma_t)^n \Sigma_0^n. \quad (8)$$

This is a **band-by-band expansion**: $\Sigma_0^0 = I$ (diagonal), Σ_0^1 is tridiagonal, Σ_0^2 is pentadiagonal (bandwidth 2), and Σ_0^n has bandwidth n . The series literally builds Σ_t^{-1} by adding one more diagonal band at each order. Figure 7 shows this in action.

3.4 F4: Spectral form — the definitive formula

Theorem 3.1 (Spectral decomposition). *Let $\Sigma_0 = UDU^\top$ be the eigendecomposition, with U orthogonal and $D = \text{diag}(d_0, \dots, d_K)$, $d_0 \geq \dots \geq d_K > 0$. Then for all $t \geq 0$:*

$$\Sigma_t^{-1} = U \text{diag}\left(\frac{1}{e^{-2t}d_0 + \Delta_t}, \dots, \frac{1}{e^{-2t}d_K + \Delta_t}\right)U^\top = \sum_{m=0}^K \frac{\mathbf{u}_m \mathbf{u}_m^\top}{e^{-2t}d_m + \Delta_t}. \quad (9)$$

Proof. Since $\Sigma_0 = UDU^\top$ and $I = UU^\top$:

$$\Sigma_t = U(e^{-2t}D + \Delta_t I)U^\top.$$

The matrix in the middle is diagonal with entries $e^{-2t}d_m + \Delta_t$. Inverting: $\Sigma_t^{-1} = U \text{diag}(1/(e^{-2t}d_m + \Delta_t))U^\top$.

The key step: adding $\Delta_t I$ to $e^{-2t}\Sigma_0$ shifts every eigenvalue by Δ_t but *leaves every eigenvector unchanged*, because I has every vector as an eigenvector. This is why Σ_t and Σ_0 share the same eigenvectors for all t . $\square$

The central insight

The eigenvectors of Σ_t^{-1} are the eigenvectors of Σ_0 , independent of t .

Only the eigenvalues change: $\lambda_m(t) = e^{-2t}d_m + \Delta_t$. As t increases, all eigenvalues converge to 1 (from above for slow modes $d_m > 1$, from below for fast modes $d_m < 1$). The matrix goes from anisotropic (elongated ellipsoid) to isotropic (sphere).

This shared-eigenvector property is a consequence of **Gaussianity** (not Markovianity) combined with **isotropic noise** ($\Delta_t \cdot I$). Any Gaussian P_0 with any covariance structure would have the same property. Markovianity determines *which* eigenvectors arise (spatially localised, sine-like modes for the AR(1) chain) and *how* the eigenvalues are distributed.

4 The Eigenbasis: Full Independence

4.1 Change of basis

Define rotated variables $\tilde{\mathbf{x}} = U^\top \mathbf{x}$, where U is the eigenvector matrix of Σ_0 . Since U is orthogonal, this is a rigid rotation—no stretching, no information loss.

Theorem 4.1 (Decoupling). *In the eigenbasis:*

(i) *The score splits into independent scalar components:*

$$\tilde{S}_m(\tilde{\mathbf{x}}, t) = -\frac{\tilde{x}_m - \tilde{\mu}_{t,m}}{e^{-2t}d_m + \Delta_t}. \quad (10)$$

Each $\tilde{S}_m$ depends only on $\tilde{x}_m$ —no coupling between modes.

(ii) *The noisy distribution factorises: $P_t(\tilde{\mathbf{x}}) = \prod_{m=0}^K \mathcal{N}(\tilde{x}_m; \tilde{\mu}_{t,m}, e^{-2t}d_m + \Delta_t)$.*

(iii) *The reverse diffusion SDE decouples into $K + 1$ independent scalar Ornstein–Uhlenbeck processes.*

Proof. For (i): $\tilde{S} = U^\top S(U\tilde{\mathbf{x}}, t) = -U^\top \Sigma_t^{-1} U(\tilde{\mathbf{x}} - \tilde{\boldsymbol{\mu}}_t)$. By Theorem 3.1, $U^\top \Sigma_t^{-1} U = \text{diag}(1/\lambda_m(t))$, giving (10).

For (ii): $U^\top \Sigma_t U = \text{diag}(\lambda_m(t))$ is diagonal, so the Gaussian factorises.

For (iii): the reverse SDE $d\mathbf{x} = (\mathbf{x} + 2S)dt + \sqrt{2}d\mathbf{W}$ in the rotated basis uses $\tilde{\mathbf{W}} = U^\top \mathbf{W}$. Since U is orthogonal and $\mathbf{W}$ has independent components, $\tilde{\mathbf{W}}$ also has independent components (rotating i.i.d. Gaussian noise gives i.i.d. Gaussian noise—the Gaussian distribution is spherically symmetric). $\square$

What this means in practice

In the original frame basis, computing the score requires solving a coupled $K+1$ -dimensional system. In the eigenbasis, it reduces to $K+1$ independent 1D problems. Each mode m is a standalone scalar denoiser with its own effective signal-to-noise ratio $\text{SNR}_m(t) = e^{-2t} d_m / \Delta_t = \gamma_t d_m$.

Slow modes (large d_m): high SNR, retain signal to large t , carry the long-range temporal structure.

Fast modes (small d_m): low SNR, overwhelmed by noise early, carry the frame-to-frame fluctuations.

This hierarchy of SNRs is the spectral fingerprint of the AR(1) dynamics on the score.

5 Numerical Study: 60 Frames

We now visualise all results for $K = 59$ (60 frames), $\alpha = 0.9$, $\sigma_\eta^2 = 1 - \alpha^2 = 0.19$, $\sigma_\infty^2 = 1$.

5.1 Σ_t^{-1} at four diffusion times

Σ_t^{-1} at four diffusion times ($K = 59, \alpha = 0.9$)

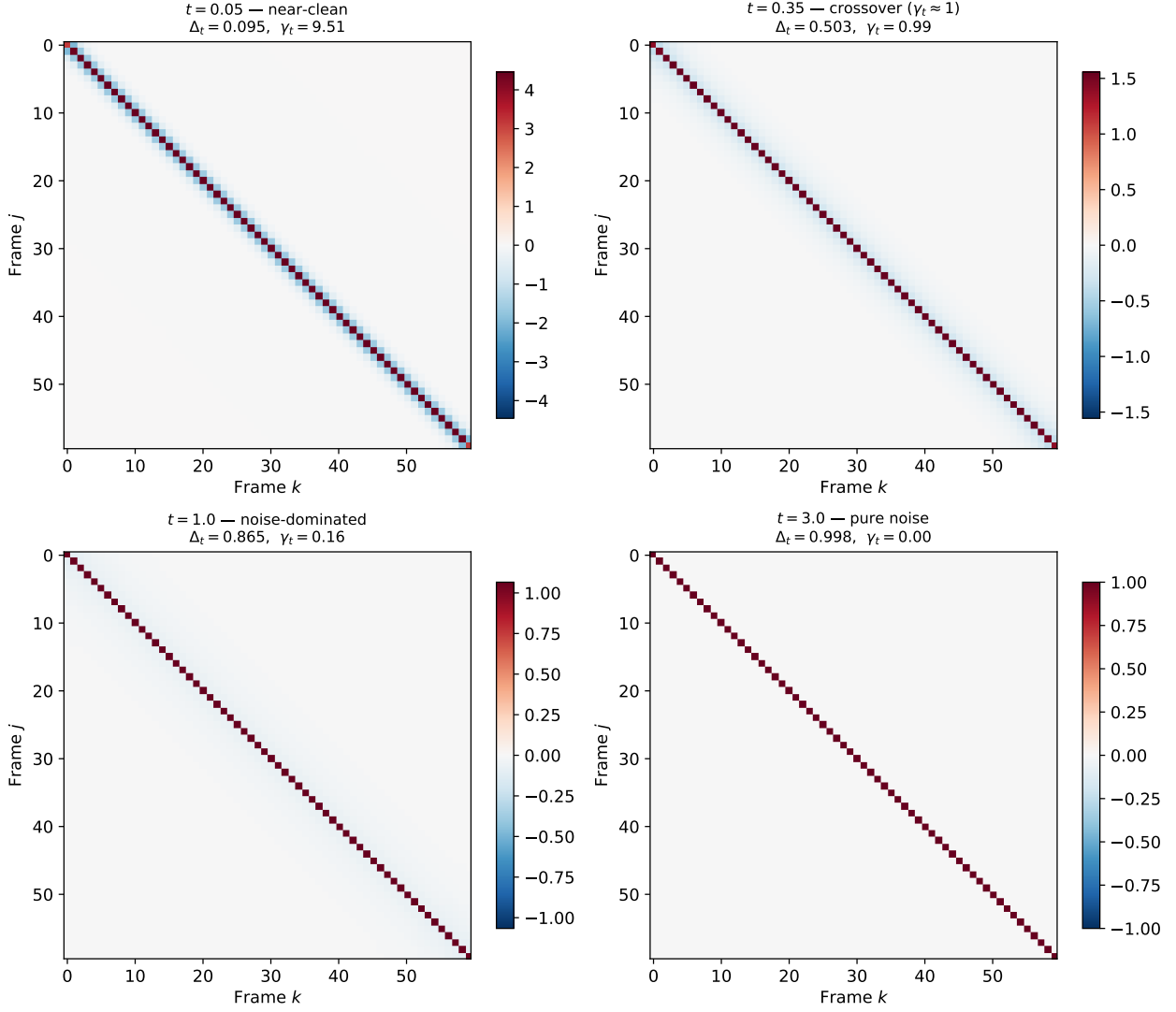


Figure 2: Σ_t^{-1} at four diffusion times. At $t = 0.05$ ($\gamma_t = 10.0$): the tridiagonal structure is visible; the dominant entries are on the three central diagonals. At $t = 0.35$ ($\gamma_t \approx 1$, the crossover): the zeros are filled but the tridiagonal band is still the strongest. At $t = 1.0$ ($\gamma_t = 0.16$): near-diagonal, with small uniform off-diagonal entries. At $t = 3.0$ ($\gamma_t = 0.003$): essentially the identity matrix.

5.2 How coupling decays with distance

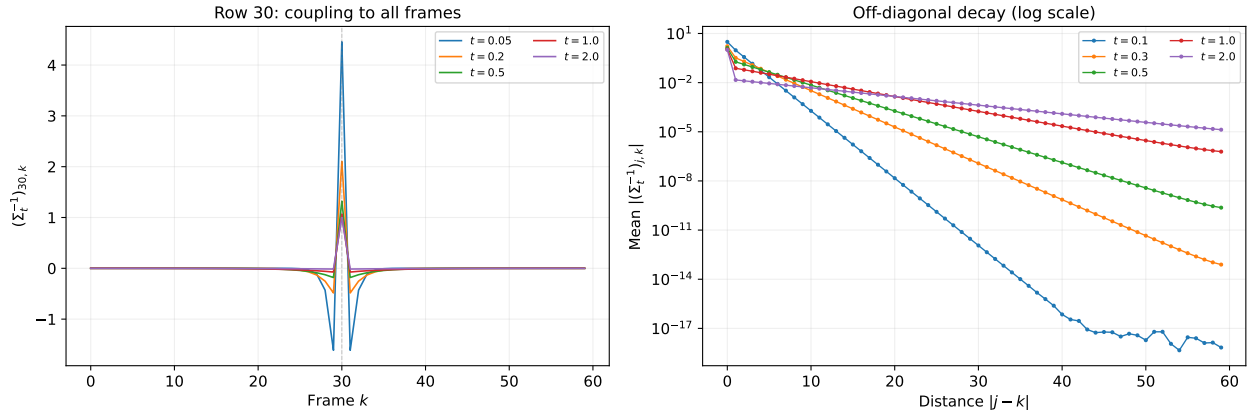


Figure 3: Left: row 30 of Σ_t^{-1} at several diffusion times. At $t = 0.05$, the coupling is sharply peaked at the nearest neighbours ($k = 29, 31$). As t grows, the peak broadens and flattens. Right: mean $|(\Sigma_t^{-1})_{j,k}|$ as a function of distance $|j - k|$ (log scale). At all $t > 0$ the decay is approximately exponential. The slope (decay rate) steepens with t , because the signal-to-noise ratio drops.

5.3 Tridiagonal weight fraction and condition number

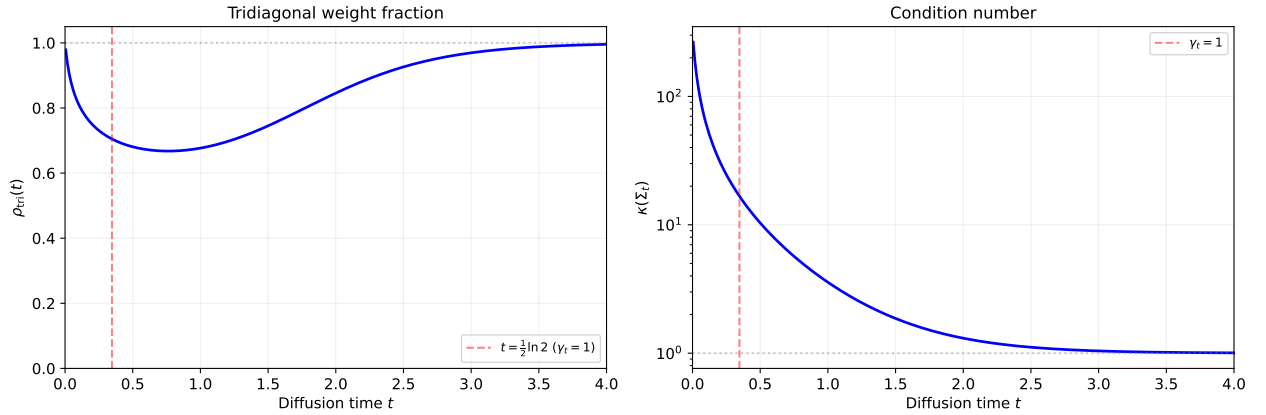


Figure 4: Left: the fraction of $\|\Sigma_t^{-1}\|_1$ (entry-wise) living on the tridiagonal band. At $t = 0$: 100%. The red dashed line marks $\gamma_t = 1$ (crossover), where $\rho_{\text{tri}} \approx 0.6$. Right: the condition number $\kappa(\Sigma_t) = \lambda_{\max}/\lambda_{\min}$ drops from ~ 315 (anisotropic signal) to 1 (isotropic noise).

5.4 Eigenvalue spectrum and eigenvectors

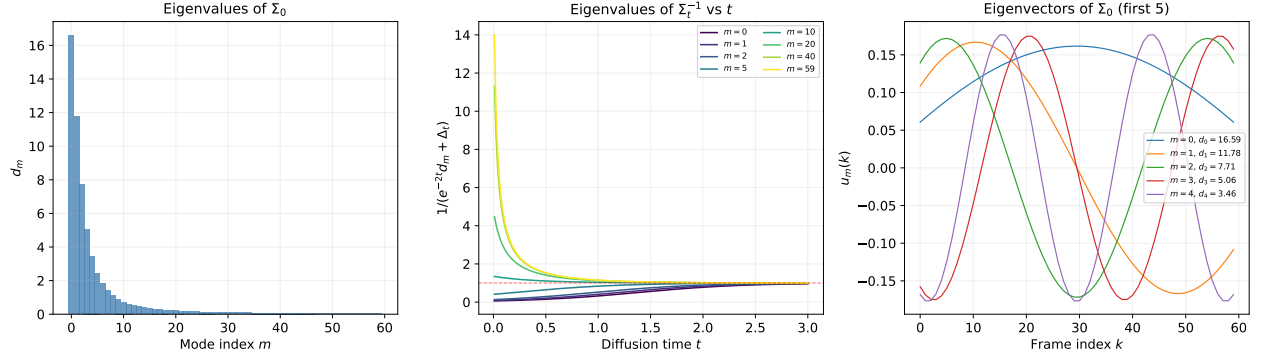


Figure 5: Left: eigenvalues of Σ_0 . The slowest mode ($m = 0$) has $d_0 \approx 16.6$; the fastest ($m = 59$) has $d_{59} \approx 0.053$. The ratio is ~ 315 , the condition number. Middle: eigenvalues of Σ_t^{-1} as functions of t for selected modes. All converge to 1. Slow modes ($m = 0$) stay large longer; fast modes ($m = 59$) collapse to 1 quickly. Right: the first five eigenvectors of Σ_0 . They are smooth, spatially localised oscillations (approximately sine functions), with mode m having m zero-crossings. These are the “Fourier-like” modes of the AR(1) chain.

5.5 Σ_t^{-1} in the eigenbasis: perfect diagonality

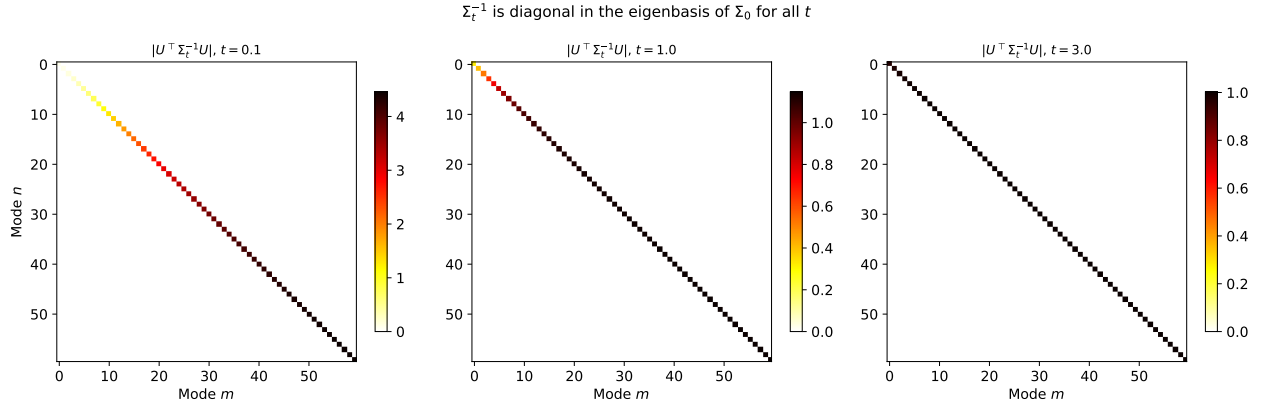


Figure 6: The matrix $|U^T \Sigma_t^{-1} U|$ at three diffusion times. It is exactly diagonal at all t , confirming that the eigenvectors of Σ_0 diagonalise Σ_t^{-1} simultaneously for all t . The diagonal entries are $1/(e^{-2t}d_m + \Delta)$.

5.6 The Neumann series: building Σ_t^{-1} band by band

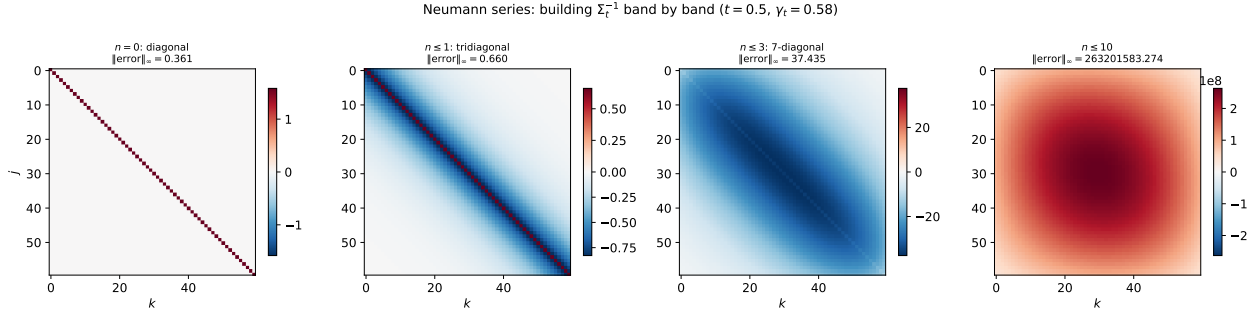


Figure 7: Partial sums of the Neumann series $\Sigma_t^{-1} = (1/\Delta_t) \sum_{n=0}^{\infty} (-\gamma_t)^n \Sigma_0^n$ at $t = 0.5$ ($\gamma_t = 0.58$). At $n = 0$: the diagonal approximation I/Δ_t . At $n \leq 1$: the tridiagonal correction is added. At $n \leq 3$: seven diagonals. By $n \leq 10$: the error is negligible. Each order adds one more band of coupling, showing how inter-frame correlations are built up perturbatively from the Markov nearest-neighbour structure.

6 Interpretation: What We Learned

6.1 The lifecycle of Markov structure in the score

As diffusion time t increases from 0 to ∞ , the precision matrix Σ_t^{-1} undergoes a continuous transformation:

1. $t \approx 0$ (**clean signal**): $\Sigma_t^{-1} \approx e^{2t} \Sigma_0^{-1}$, tridiagonal. The score couples each frame only to its nearest neighbours. Full Markov structure.
2. **Small t** ($\gamma_t \gg 1$): the zeros begin to fill in, but the tridiagonal entries remain dominant. The beyond-tridiagonal entries decay exponentially with distance from the diagonal. The Markov “fingerprint” is strong.
3. **Crossover** $\gamma_t = 1$ ($t \approx 0.35$): signal and noise contribute equally. The tridiagonal band still carries $\sim 60\%$ of the total weight, but long-range couplings are no longer negligible.
4. **Large t** ($\gamma_t \ll 1$): $\Sigma_t^{-1} \approx I/\Delta_t$. Each frame’s score depends only on itself. All inter-frame coupling has been erased. The score simply pushes each x_k back toward $\mu_{t,k}$.

6.2 Separating the roles of Gaussianity and Markovianity

Property	Requires
Σ_0^{-1} is tridiagonal (sparse)	Markov
$P_t(\mathbf{x})$ is Gaussian; score is linear in $\mathbf{x}$	Gaussian
Σ_t, Σ_0 share eigenvectors for all t	Gaussian + isotropic noise
Score decouples in eigenbasis	Gaussian + isotropic noise
Eigenvectors are sine-like / spatially localised	Markov (tridiagonal Σ_0^{-1})
$d_m = \sigma_\eta^2 / (1 + \alpha^2 - 2\alpha \cos \theta_m)$	AR(1) dynamics
$ (\Sigma_t^{-1})_{jk} $ decays with $ j - k $	Markov (indirect)

Gaussianity provides the existence of a fully decoupling eigenbasis. **Markovianity** provides the specific structure of that basis (what the eigenvectors look like, what the eigenvalue spectrum is) and is responsible for the spatial localisation of couplings in Σ_t^{-1} .

If we broke Gaussianity (e.g. mixture of Gaussians), the eigenbasis decoupling would fail: the score would become nonlinear in $\mathbf{x}$, and no rotation could make the components independent. If we broke Markovianity (e.g. a Gaussian with long-range correlations), the decoupling would still work, but the eigenvectors and eigenvalues would change.

7 Looking Ahead

Following the research programme outlined by Mézard:

1. **Enrich P_0** : give P_0 a non-trivial covariance structure within each frame (multivariate frames with $a_k \in \mathbb{R}^d$), to disentangle intra-frame from inter-frame effects.
2. **Break Gaussianity**: replace P_0 with a mixture of Gaussians. The score becomes nonlinear, the eigenbasis trick fails, and the question becomes: does the *Markov* structure (nearest-neighbour dominance) survive in the score even when the Gaussian decoupling does not?
3. **Quantify the Markov residual**: define sharper indicators (beyond ρ_{tri}) that measure how much of the score's structure is attributable to Markovianity at each diffusion time.
4. **Connect to the Kalman smoother**: the backward pass of the Kalman smoother computes $\langle a_k \rangle_{\mathbf{a}|\mathbf{x}}$ in $O(K)$ time by propagating information from frame $k+1$ to frame k . This propagator is the Markov structure of the score made algorithmic.

8 Notation

Symbol	Meaning
K	Number of AR(1) steps; $K+1$ frames total
$t \geq 0$	Diffusion time
$\alpha \in (-1, 1)$	AR(1) contraction coefficient
σ_η^2	Innovation variance; $\sigma_\infty^2 = \sigma_\eta^2 / (1 - \alpha^2)$ stationary variance
$\Delta_t = 1 - e^{-2t}$	OU noise variance
$\gamma_t = e^{-2t} / \Delta_t$	Signal-to-noise ratio
$\Sigma_0, \Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$	Clean / noisy covariance
$d_m, \mathbf{u}_m, U$	Eigenvalues, eigenvectors, eigenvector matrix of Σ_0
$\lambda_m(t) = e^{-2t}d_m + \Delta_t$	Eigenvalue of Σ_t
$\tilde{\mathbf{x}} = U^\top \mathbf{x}$	Trajectory in the eigenbasis
$S(\mathbf{x}, t) = -\Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t)$	Joint score
$\rho_{\text{tri}}(t)$	Tridiagonal weight fraction of Σ_t^{-1}